

RAJESH VAYYALA

PRINCIPAL DATA ARCHITECT

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EXECUTIVE SUMMARY

Seasoned Principal Data Architect and Engineering Leader with over 10 years of experience designing and modernizing enterprise data ecosystems across financial services and investment domains. Proven expertise in data architecture, integration, and governance, delivering scalable platforms on Microsoft Fabric, Snowflake, and Azure. Adept at leading cross-functional teams to unify internal and third-party data sources, automate ELT pipelines, and enable analytics and BI solutions that drive strategic insights and operational excellence. Specialize in translating business objectives into governed, analytics-ready, and AI-enabled data structures that power critical reporting and decision-making.

Core Competencies: Azure Cloud Architecture — Petabyte-Scale Data Platforms — Enterprise Data Strategy and Governance — Real-time Analytics — Regulatory Compliance — Data Analytics and Insights Leadership — KPI Design and Visualization — Financial Services Analytics — Stakeholder and Executive Engagement

PROFESSIONAL EXPERIENCE

PRA Group Inc

Oct 2024 - Oct 2025

Principal Data Architect

- **Enterprise Azure Data Platform Modernization:** Led a cloud modernization initiative serving multiple business units across regions, modernizing data architecture and governance within Microsoft Fabric to enable unified analytics and trusted insights.
 - Led enterprise-wide data architecture modernization initiatives aligning with corporate strategy and analytics needs across finance, operations, and investment domains.
 - Designed and implemented Medallion-style architectures (Bronze/Silver/Gold) using Microsoft Fabric, Snowflake, and Azure Synapse, improving scalability, data quality, and analytics readiness.
 - Designed and built end-to-end data solutions on Microsoft Fabric, integrating, data ingestion, transformation, storage, and reporting layers.
 - Developed and optimized lakehouse and data warehouse architectures in Azure Fabric to support enterprise analytics and BI workloads.
 - Implemented data pipelines using Dataflows Gen2 and Data Factory within Fabric for efficient and scalable data movement.
 - Improved data processing and analytics performance by **30–35%** through a modern fabric lakehouse approach.
 - Created data models and semantic layers in Fabric for consistent and governed reporting across business functions.
 - Implemented a company-wide data governance framework using Erwin Data Mart, ensuring consistency and accessibility across business domains.
 - Built interactive Power BI reports and dashboards connected to Fabric datasets, enabling real-time insights for business users.
- **Global Hybrid Cloud Data Platform Integration:** Led a large-scale, petabyte-level migration of legacy enterprise systems to a modern Azure-based hybrid cloud platform, coordinating 15+ cross-functional teams across global business units. Established DataOps and DevOps best practices while ensuring compliance with international data regulations and enterprise security policies.
 - Reduced storage costs by **20%** (\$2M+ annual savings) and data ingestion time by **25%**.
 - Improved data accuracy by **30%** through advanced quality frameworks.
 - Architected real-time pipelines processing 10TB+ daily for 500,000+ global shipments.
- **Azure Synapse Analytics Modernization:** Architected and modernized the enterprise data warehouse ecosystem managing over 100TB of business-critical data. Designed a high-performance analytics architecture integrating Azure Synapse, Snowflake, SQL Server, and Data Factory, improving data availability, reliability, and speed for enterprise reporting.
 - Accelerated query performance by **30%** and reduced data loading from hours to minutes.
 - Implemented unified data pipelines using Snowflake, SQL Server, and Azure Synapse, improving data freshness and analytical accuracy by 40%.
 - Performed in-depth data analysis using Snowflake SQL to identify business trends, data inconsistencies, and optimization opportunities.
 - Translated analytical findings into strategic recommendations for executive stakeholders, influencing data-driven business decisions across multiple portfolios.
 - Designed and developed interactive dashboards in Power BI and TIBCO Spotfire transforming complex datasets into actionable insights for executive and operational decision-making.

FedEx Services

Jun 2021 - Oct 2024

Data Architect

First Horizon Bank

Jun 2019 - Jun 2021

Data Architect Modeler

- **Customer Data Platform Modernization (Cloudera Hadoop CDP):** Led the design and implementation of a large-scale Customer Data Platform initiative, consolidating enterprise data assets into a unified analytics environment. Defined strategic data layers and governance standards across multiple legacy and modern systems.
 - Integrated 80+ key datasets from diverse business domains into a centralized platform, supporting 200TB+ data growth.
 - Delivered **20% faster customer insights** and achieved annual operational savings of over **\$1M** through optimized data pipelines and automation.
 - Established foundational CDP architecture adopted across entire financial services division.
- **Enterprise Data Quality & OLAP Transformation:** Architected an enterprise-wide data quality and analytics framework to strengthen accuracy, consistency, and trust in reporting and performance management. Designed OLAP models and STAR schemas optimized for large-scale analytical workloads.
 - Improved data accuracy by **90%** across critical business metrics and KPIs.
 - Reduced report generation time by over **80%**, enabling near real-time visibility for executive dashboards.
 - Enabled regulatory and risk analytics automation, enhancing decision-making and compliance efficiency across business lines.
 - Built enterprise and dimensional data models in Erwin, improving reporting efficiency by **20%** and ensuring consistent KPI definitions across business units.
 - Automated recurring analytical pipelines in Python and SQL, reducing manual effort and improving report turnaround time by **40%**.

IS Tech INC

Dec 2015 - Jun 2019

Data Modeler/Metadata Analyst *Clients: J.B.Hunt, Wells Fargo & Co., Anthem BCBS*

- **Enterprise Data Governance Platform:** Developed comprehensive data models for 3 Fortune 500 clients managing 500+ entities and 10,000+ attributes. Built centralized Erwin metadata repository with 2,000+ business rules serving 100+ development teams across healthcare, financial, and logistics domains.
 - Reduced data model maintenance by **15%**, saving 200+ developer hours monthly.
 - Implemented Master Data Management, processing millions of records daily.
 - Standardized modeling practices across \$100B+ combined client revenues.
- **Data Lifecycle Management Framework:** Architected retention framework managing 50TB+ regulated data across multiple jurisdictions for 100+ systems. Implemented automated archival processes ensuring GDPR, SOX, and HIPAA compliance while protecting 50M+ customer records.
 - Achieved **40% storage cost reduction** (\$1M+ annual savings).
 - Maintained **100% regulatory compliance** across all au-

dits.

- Streamlined purging processes reducing operational overhead by **30%**.

Adroix Corp

Jul 2015 - Dec 2015

Computer System Analyst

- Analyzed 15+ interconnected legacy systems identifying optimization opportunities across 200+ data flows.
- Developed enterprise modeling standards adopted by 50+ analysts supporting 20+ clients generating \$500M+ revenues.
- Reduced data redundancy by **30%** saving 10TB storage.
- Improved query performance by **25%** affecting 1,000+ daily reports.
- Created reusable templates reducing development time by **40%**.

EDUCATION

MS Computer Science

2015

Fairleigh Dickinson University, NJ — GPA: 3.5/4.0

- Academic Excellence Scholarship Recipient (Full Tuition)
- Graduated with Distinction — Focus: Big Data & Distributed Systems

CERTIFICATIONS

- Microsoft Certified: Azure Data Fundamentals (DP-900) - Validates cloud data concepts expertise
- Microsoft Certified: Azure Fundamentals (AZ-900) - Demonstrates comprehensive Azure knowledge
- Certified SAFe® 5 Practitioner - Agile methodology for enterprise-scale implementations
- Big Data Expert Certificate - Edureka, 200+ hours of hands-on training
- FedEx Business Essentials Certificate - Executive leadership development program
- FedEx Management Exploration Program Graduate - Selected from top 5% of employees

TECHNICAL EXPERTISE

- **Cloud & Big Data:** Azure (Synapse Analytics, Data Factory, Data Lake Gen2, Blob Storage, Purview, SQL Database), Cloudera Hadoop CDP, Cosmos DB
- **Databases:** Oracle 12c/19c, SQL Server 2016-2022, Teradata, IBM DB2, PostgreSQL
- **Data Modeling & Design:** Erwin Data Modeler 9.x, PowerDesigner, ER Studio, Dimensional Modeling, Data Vault 2.0
- **Analytics & BI:** Tableau Server, Power BI, Spotfire, Hive, Spark SQL, Tibco Data Virtualization
- **Data Governance:** Collibra EDGE, Ab Initio EMH, Azure Purview, Informatica MDM
- **Programming & Tools:** SQL, Python, Azure DevOps, Git, JIRA, Confluence
- **Methodologies:** DataOps, DevOps, Agile/SAFe, CI/CD, Data Mesh Architecture